Chapter - 4

Fibrous Proteins

Lecture 7

Contents:

- Overview
- Collagen:
  - Types of collagen
  - Structure of collagen
  - Biosynthesis of collagen

I. OVERVIEW

- **Collagen and elastin are examples of common, well-characterized fibrous proteins that serve structural functions in the body.**

- **For example, collagen and elastin are found as components of skin, connective tissue, blood vessel walls, and sclera and cornea of the eye.**

- Each fibrous protein exhibits special mechanical properties, resulting from its unique structure, which are obtained by combining specific amino acids into regular, secondary structural elements.

- This is in contrast to globular proteins, whose shapes are the result of complex interactions between secondary, tertiary, and sometimes, quaternary structural elements.

II. COLLAGEN

- **Collagen is the most abundant protein in the human body.**

- **A typical collagen molecule is a long, rigid structure in which three polypeptides "α-chains" are wound around one another in a rope-like triple-helix *(Figure 4.1).***

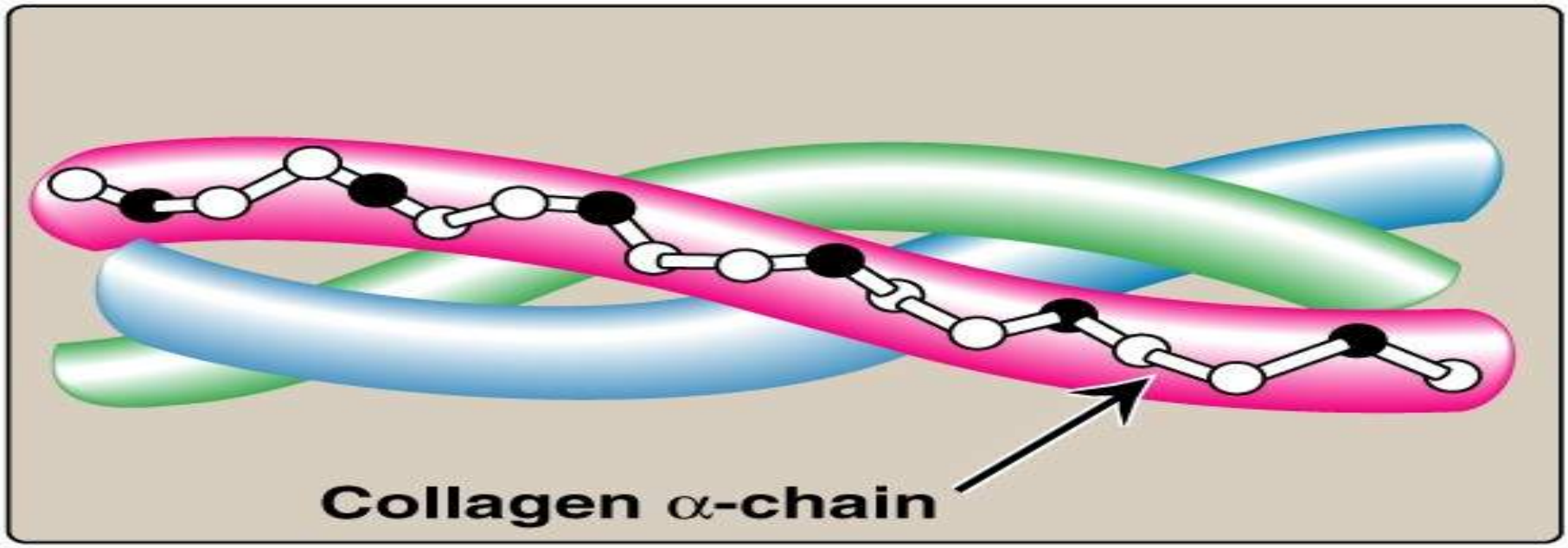


Figure 4.1
Triple-stranded helix of collagen.

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- **Although collagen molecules are found throughout the body, their types and organization are dictated by the structural role collagen plays in a particular organ.**

- **In some tissues, collagen may be dispersed as a gel → support to the structure, as *in the extracellular matrix* or the *vitreous humor of the eye*.**

- In other tissues, collagen may be bundled in tight, parallel fibers → great strength, as in tendons.

- In the cornea of the eye, collagen is stacked → transmit light with a minimum of scattering.

- Collagen of bone occurs as fibers arranged at an angle to each other → resist mechanical shear from any direction.

A. Types of collagen

- The **collagen superfamily** of proteins includes more than **20 collagen types**, and proteins that have collagen-like domains.

- **The three polypeptide α-chains** are ***held together by*** **hydrogen bonds** between the chains.

- **Variations in the amino acid sequence of the α-chains → structural components that are about the same size *(approximately 1000 amino acids long),* but with slightly different properties.**

- **These α-chains are combined to form the various types of collagen found in the tissues.**

- the most common collagen:
  - Type I, contains two chains called αI and one chain called α2 ($\alpha 1_2 \alpha 2$)
  - type II collagen contains three αI chains ($\alpha 1_3$).

- The collagens can be organized into three groups, based on their location and functions in the body *(Figure 4.2).*

TYPE	TISSUE DISTRIBUTION
	Fibril-forming
I	Skin, bone, tendon, blood vessels, cornea
II	Cartilage, intervertebral disk, vitreous body
III	Blood vessels, fetal skin
	Network-forming
IV	Basement membrane
VII	Beneath stratified squamous epithelia
	Fibril-associated
IX	Cartilage
XII	Tendon, ligaments, some other tissues

Figure 4.2
The most abundant types of collagen.

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:Fibril-forming collagens .1

- **Types I, II, and III are the fibrillar collagens, and have the rope-like structure described before for a typical collagen molecule.**

- **In the electron microscope, these linear polymers of fibrils have characteristic banding patterns → reflecting the regular staggered packing of the individual collagen molecules in the fibril *(Figure 4.3).***

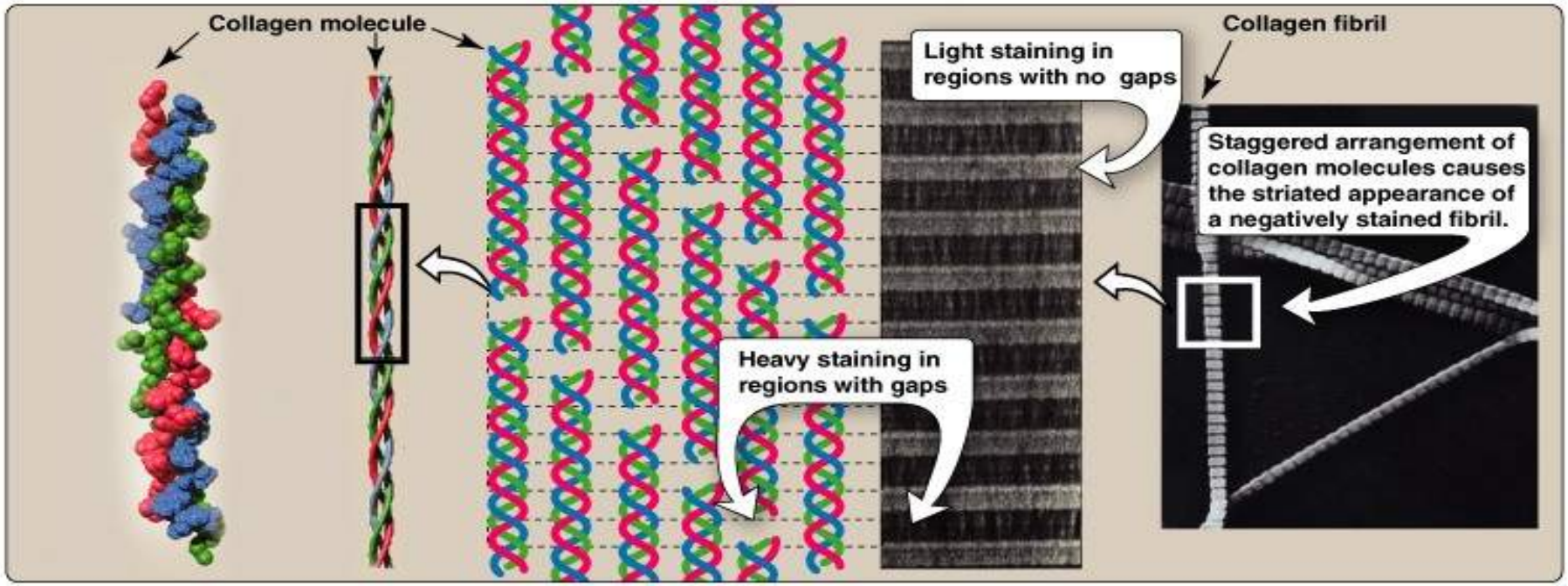


Figure 4.3

Collagen fibrils at right have a characteristic banding pattern, reflecting the regularly staggered packing of the individual collagen molecules in the fibril.

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- **Type I collagen fibers are found in supporting elements of high tensile strength (e.g. tendon and cornea).**
- **Fibers formed from type II collagen molecules are restricted to cartilaginous structures.**
- **Fibrils derived from type III collagen are prevalent in more distensible tissues, such as blood vessels.**

2. Network-forming collagens:

- **Types IV and VII form a three-dimensional mesh, rather than distinct fibrils *(Figure 4.4).***
- **For example, type IV molecules assemble into a sheet or meshwork that constitutes a major part of basement membranes.**
- ***[Basement membranes are thin, sheet-like structures that provide mechanical support for adjacent cells, and function as a semipermeable filtration barrier for macromolecules in organs such as the kidney and the lung.]***

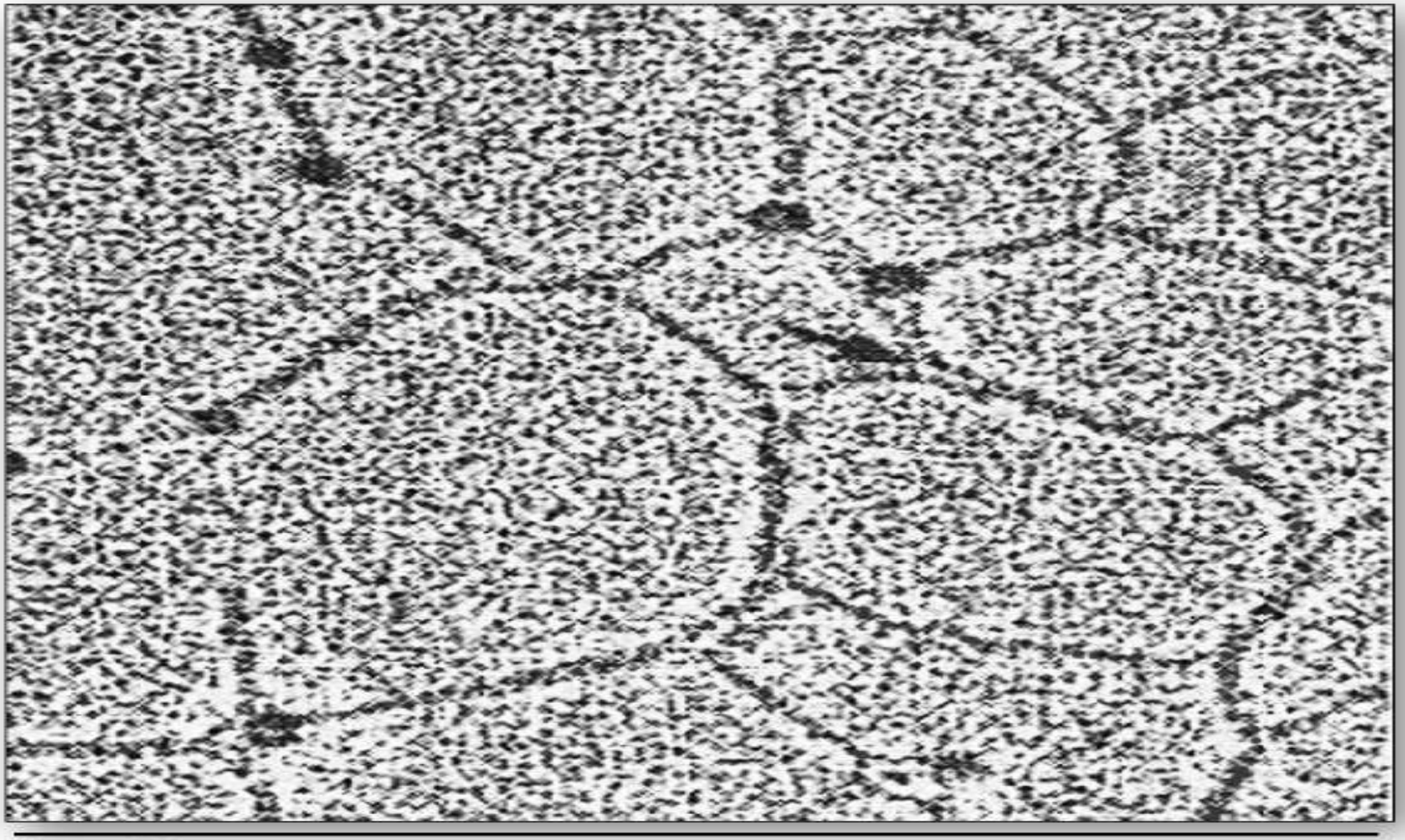


Figure 4.4

Electron micrograph of a polygonal network formed by association of collagen type IV monomers.

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:Fibril-associated collagens .3

- **Types IX and XII bind to the surface of collagen fibrils, linking these fibrils to one another and to other components in the extracellular matrix *(Figure 4.2).***

TYPE	TISSUE DISTRIBUTION
	Fibril-forming
I	Skin, bone, tendon, blood vessels, cornea
II	Cartilage, intervertebral disk, vitreous body
III	Blood vessels, fetal skin
	Network-forming
IV	Basement membrane
VII	Beneath stratified squamous epithelia
	Fibril-associated
IX	Cartilage
XII	Tendon, ligaments, some other tissues

Figure 4.2
The most abundant types of collagen.

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B. Structure of collagen

1. Amino acid sequence:

- **Collagen is rich in proline and glycine, both of which are important in the formation of the triple-stranded helix.**

- **Proline → facilitates the formation of the helical conformation of each α-chain because its ring structure → "kinks" in the peptide chain.**

- **Glycine, the smallest amino acid, is found in every third position of the polypeptide chain.**

- **It fits into the restricted spaces where the three chains of the helix come together.**

- The glycine residues are part of a repeating sequence. —Gly—X—Y—, where X is frequently proline and Y is often hydroxyproline or hydroxylysine *(Figure 4.5).*

- Most of the α.- chain can be regarded as a polytripeptide whose sequence can be represented as (—Gly—X—Y—) $_{333}$

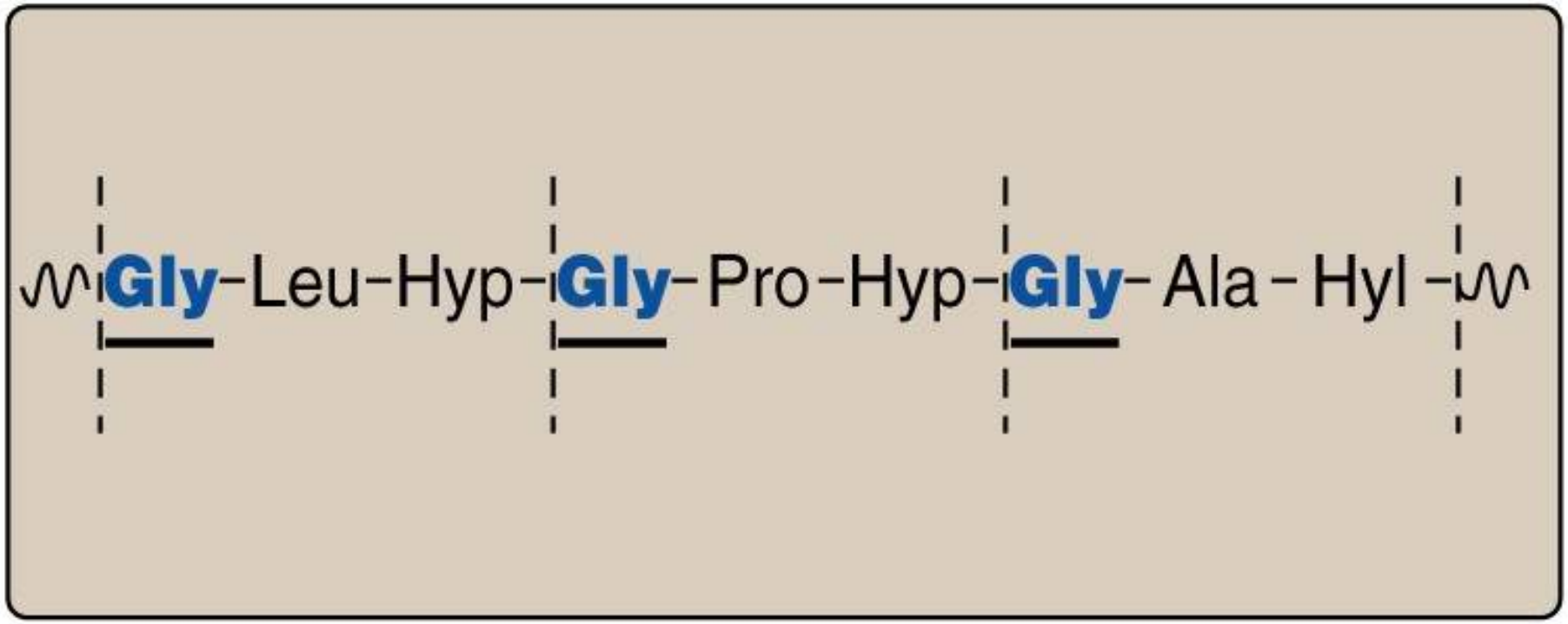


Figure 4.5

Amino acid sequence of a portion of the α1-chain of collagen. [Note: Hyp is hydroxyproline and Hyl is hydroxylysine.]

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:Triple-helical structure .2

- **Unlike most globular proteins that are folded into compact structures, collagen, a fibrous protein, has an elongated, triple-helical structure that places many of its amino acid side chains on the surface of the triple-helical molecule.**

- ***[This allows bond formation between the exposed R-groups of neighboring collagen monomers → aggregation into long fibers]***

:Hydroxyproline and hydroxylysine .3

- **Collagen contains hydroxyproline (hyp) and hydroxylysine (hyl), which are not present in most other proteins.**

- **These residues result from the hydroxylation of some of the proline and lysine residues after their incorporation into polypeptide chains**

(Figure 4.6).

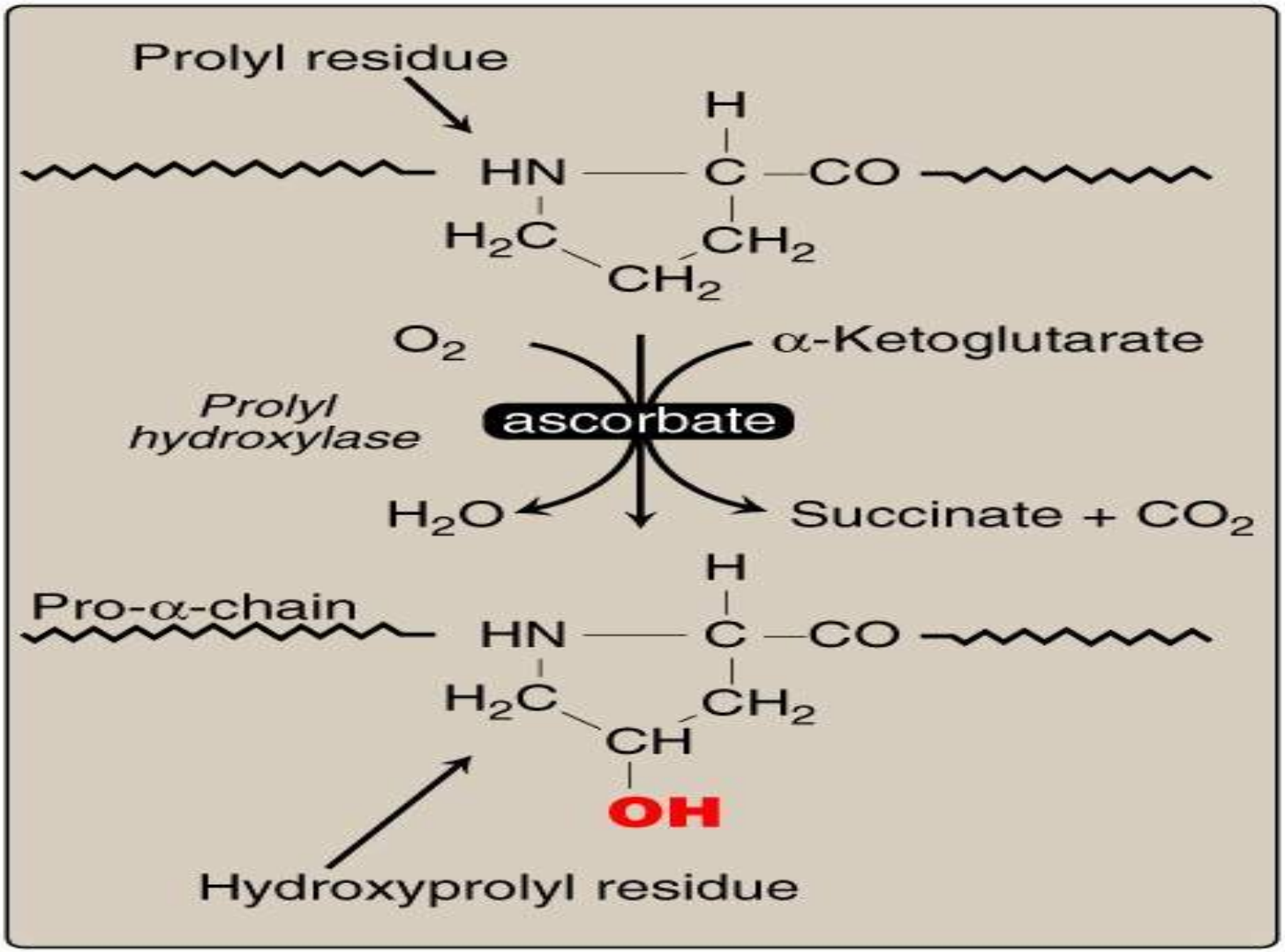


Figure 4.6
Hydroxylation of prolyl residues of pro-α-chains of collagen by *prolyl hydroxylase.*

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- The hydroxylation is, thus, an example of posttranslational modification .

- Hydroxyproline is important in stabilizing the triple-helical structure of collagen because it maximizes interchain hydrogen bond formation.

:Glycosylation .4

- The **hydroxyl group of the hydroxylysine residues of collagen** may be enzymatically **glycosylated**.

- **Most commonly**, **glucose** and **galactose** are **sequentially attached to the polypeptide chain** prior to triple-helix formation ***(Figure 4.7)***.

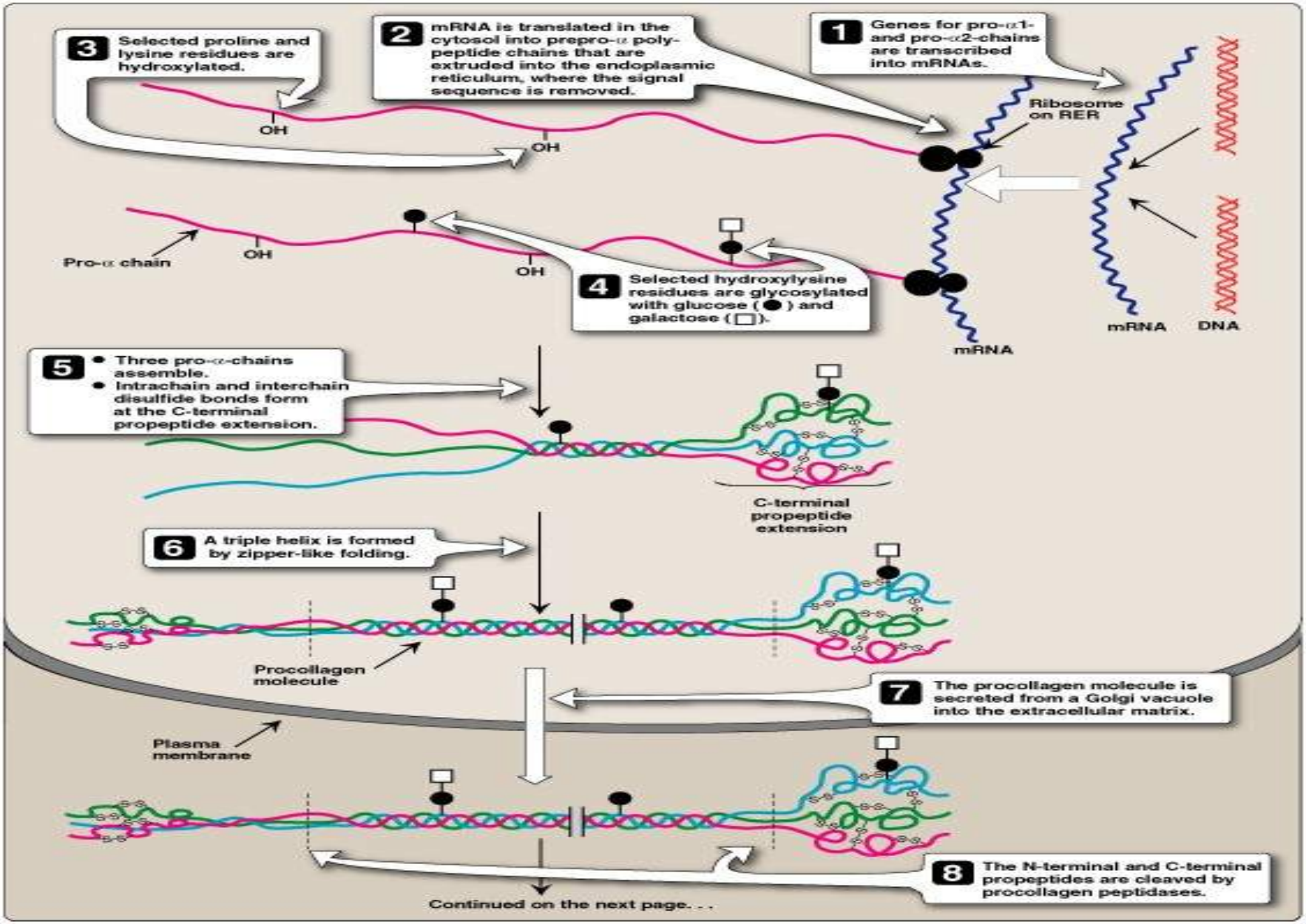


Figure 4.7
Formation of a collagen fibril. (*Continued on the next page*)

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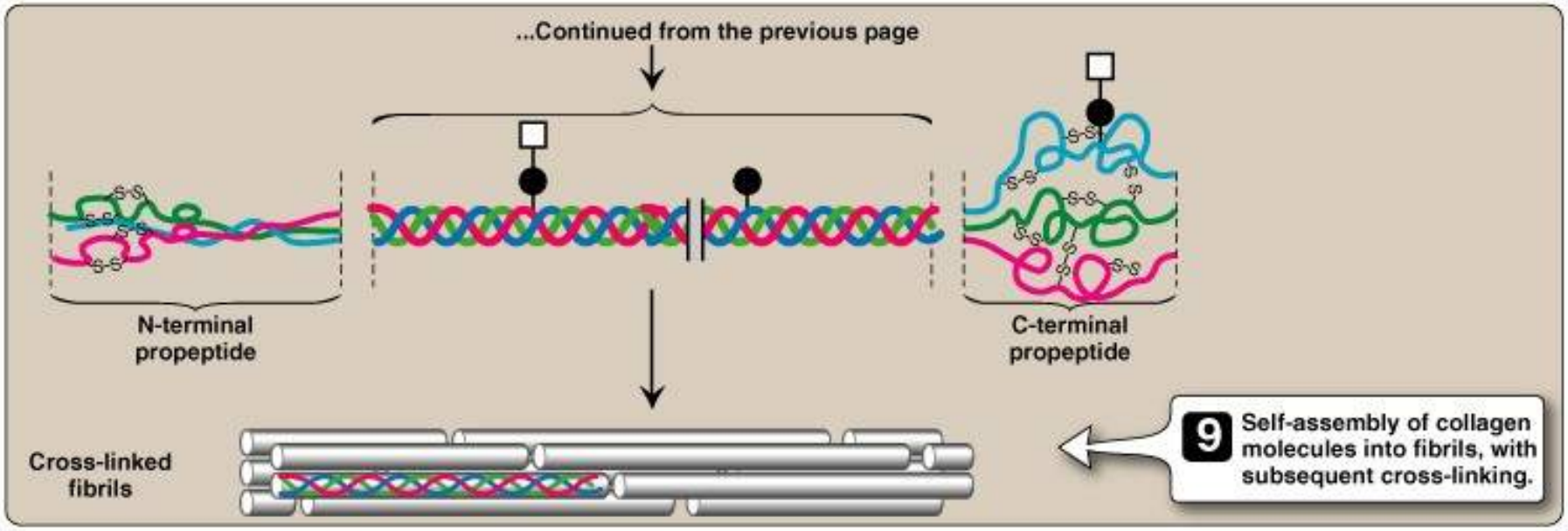


Figure 4.7

Formation of a collagen fibril. (Continued from the previous page)

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C. Biosynthesis of collagen

- The polypeptide precursors of the collagen molecule are formed in fibroblasts (or in the related osteoblasts of bone and chondroblasts of cartilage), and are secreted into the extracellular matrix.
- After enzymic modification, the mature collagen monomers aggregate and become cross-linked → collagen fibrils.

1. Formation of pro- α-chains:

- **Collagen is one of many proteins that normally function outside of cells.**
- **Like most proteins produced for export, the newly synthesized polypeptide precursors of α-chains contain a special amino acid sequence at their N-terminal ends.**
- **This acts as a signal that the polypeptide being synthesized is destined to leave the cell.**

- The signal sequence:
  - facilitates the binding of ribosomes to the rough endoplasmic reticulum (RER)
  - directs the passage of the polypeptide chain into the cisternae of the RER.
  - is rapidly cleaved in the endoplasmic reticulum → precursor of collagen called a pro-α-chain *(Figure 4.7).*

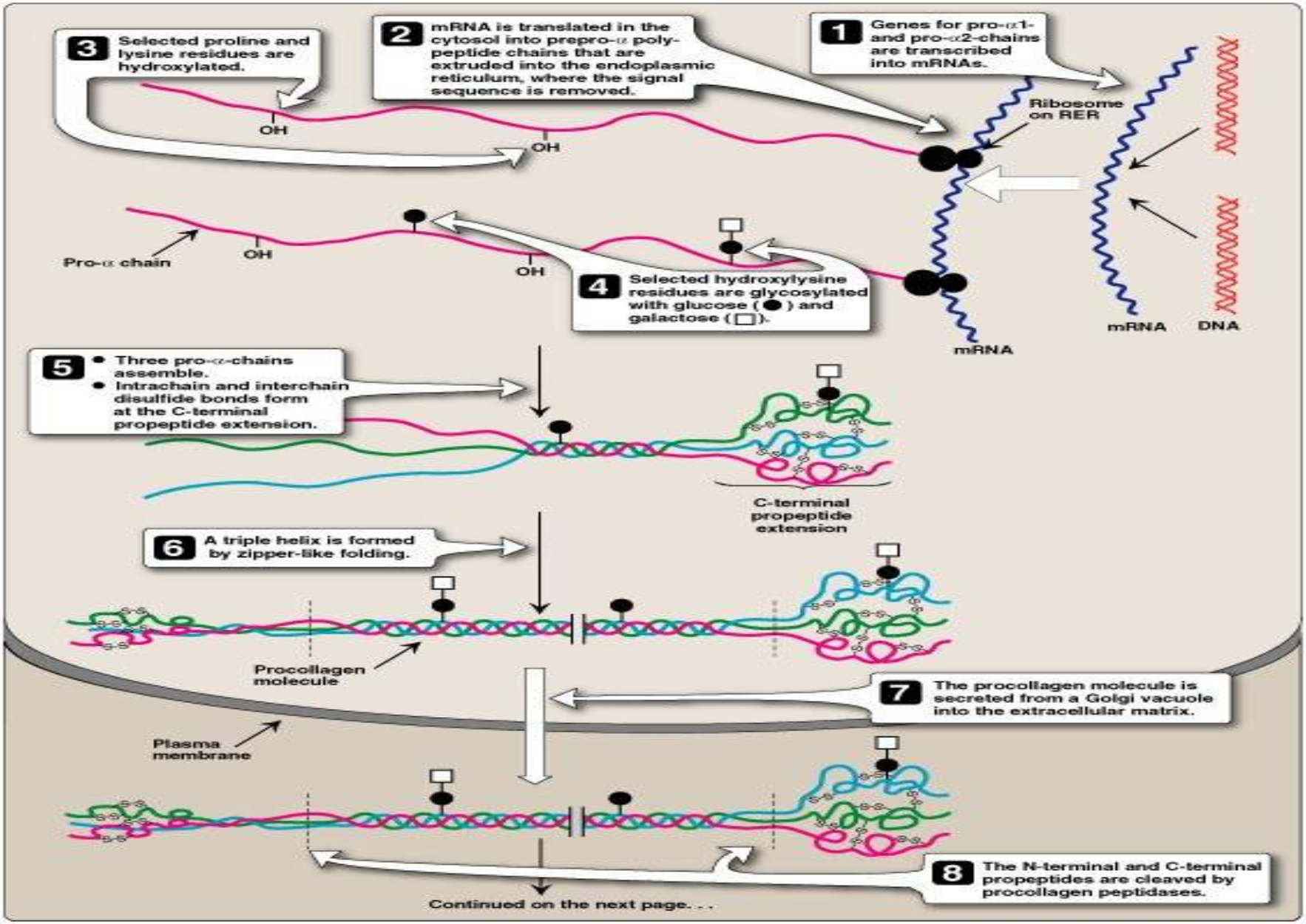


Figure 4.7
Formation of a collagen fibril. (*Cont ed on the next page*)

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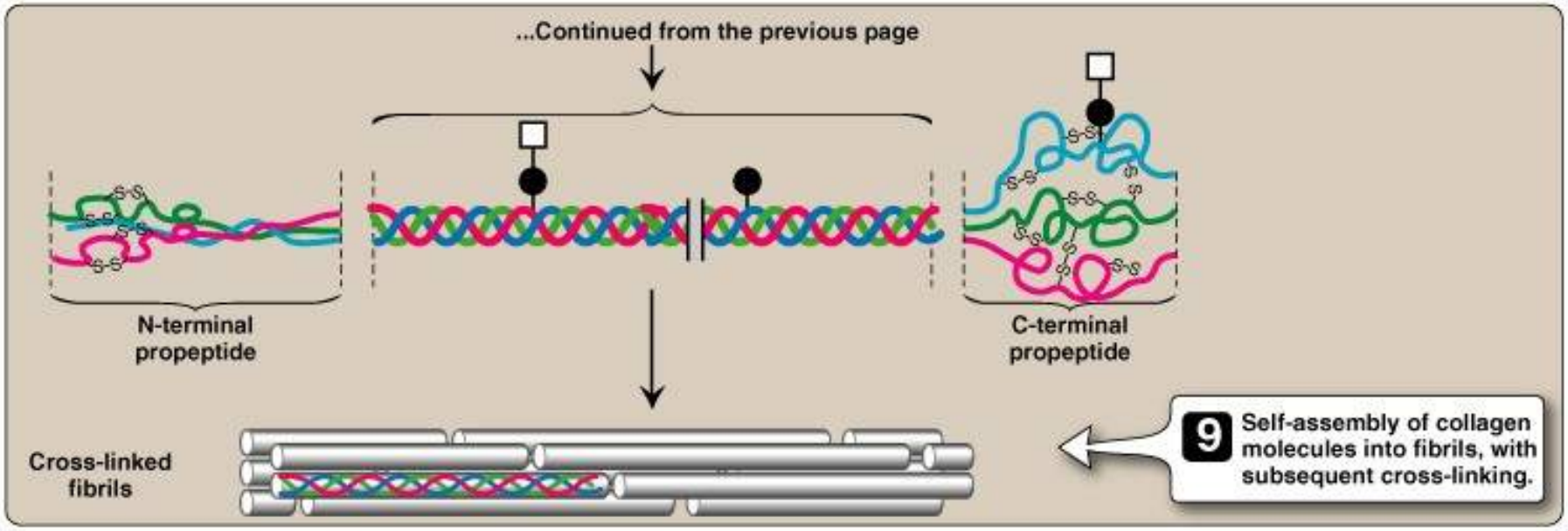


Figure 4.7

Formation of a collagen fibril. (Continued from the previous page)

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:Hydroxylation .2

- The pro- α-chains are processed by a number of enzymic steps within the lumen of the RER while the polypeptides are still being synthesized *(Figure 4.7).*
- Proline and lysine residues found in the Y-position of the —Gly—X—Y— sequence can be hydroxylated → hydroxyproline and hydroxylysine residues.

- These hydroxylation reactions require molecular oxygen and the reducing agent vitamin C
- the hydroxylating enzymes, *prolyl hydroxylase* and *lysyl hydroxylase*, are unable to function without vit. C

(Figure 4.6).

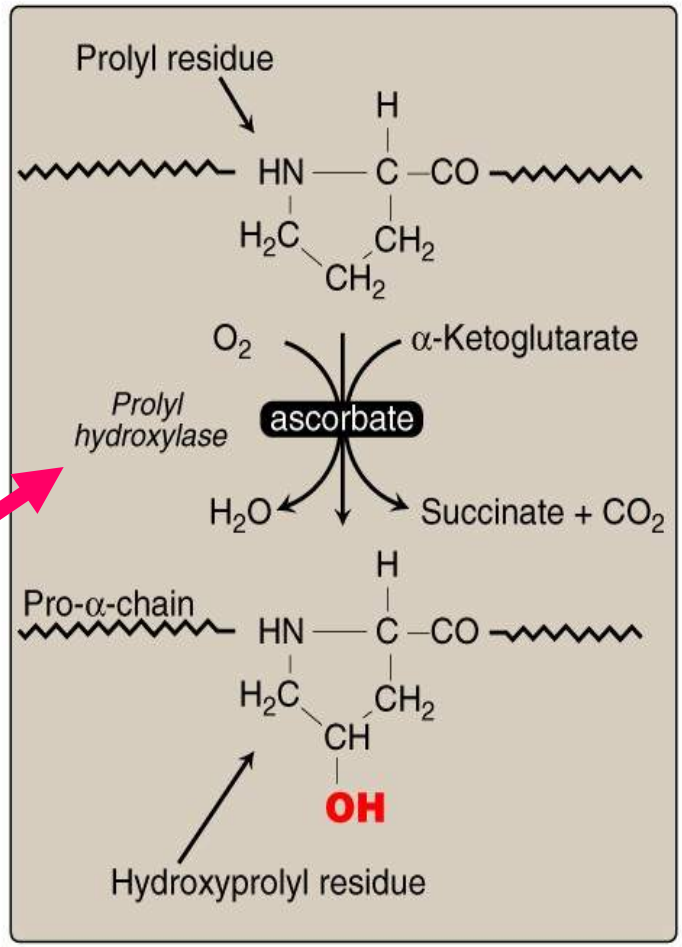


Figure 4.6
Hydroxylation of prolyl residues of pro-α-chains of collagen by *prolyl hydroxylase.*

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- In the case of vit.C deficiency *(therefore, a lack of prolyl and lysyl hydroxylation)*, collagen fibers cannot be cross-linked, greatly → ⇊ the tensile strength of the assembled fiber.

- Vit.C deficiency → disease known as scurvy.

- Patients with vit.C deficiency often show bruises on the limbs as a result of subcutaneous extravasation of blood (capillary fragility) *(Figure 4.8)*



Figure 4.8

The legs of a 46-year-old man with scurvy.

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:Glycosylation .3

- Some hydroxylysine residues are modified by glycosylation with glucose or glucosyl-galactose *(Figure 4.7)*.

:Assembly and secretion .4

- **After hydroxylation and glycosylation, pro-α-chains → form procollagen ,**
  a precursor of collagen that has a central region of triple helix flanked by the non-helical amino- and carboxyl-terminal extensions called propeptides.

(Figure 4.7).

- The formation of procollagen begins with formation of interchain disulfide bonds between the C-terminal extensions of the pro-α- chains.

- This brings the three α-chains into an alignment favorable for helix formation.

- The procollagen molecules are translocated to the Golgi apparatus, where they are packaged in secretory vesicles.

- The vesicles fuse with the cell membrane → release of procollagen molecules into the extracellular space.

5. Extracellular cleavage of procollagen molecules:

- After their release, the procollagen molecules are *cleaved by N - and C – pro - collagen peptidases* → remove the terminal propeptides → releasing triple-helical collagen molecules.

:Formation of collagen fibrils .6

- **Individual collagen molecules spontaneously associate → form fibrils.**
- **They form an ordered, overlap ping, parallel array, with adjacent collagen molecules arranged in a staggered pattern.**
- **Each collagen molecule overlaps its neighbor by three-quarters of its length.**

(Figure 4.7).

:Cross-link formation .7

- The fibrillar array of collagen molecules serves as a substrate for *lysyl oxidase*.
- This *extracellular enzyme* oxidatively deaminates some of the lysyl and hydroxylysyl residues in collagen.

- **The reactive aldehydes that result (*allysine* and *hydroxyallysine*) → condense with lysyl or hydroxylysyl residues in neighboring collagen molecules → form covalent cross-links *(Figure 4.9).***
- ***This cross-linking is essential for achieving the tensile strength necessary for the proper functioning of connective tissue.***
- ***Any mutation that interferes with the ability of collagen to form cross-linked fibrils → affects the stability of the collagen].***

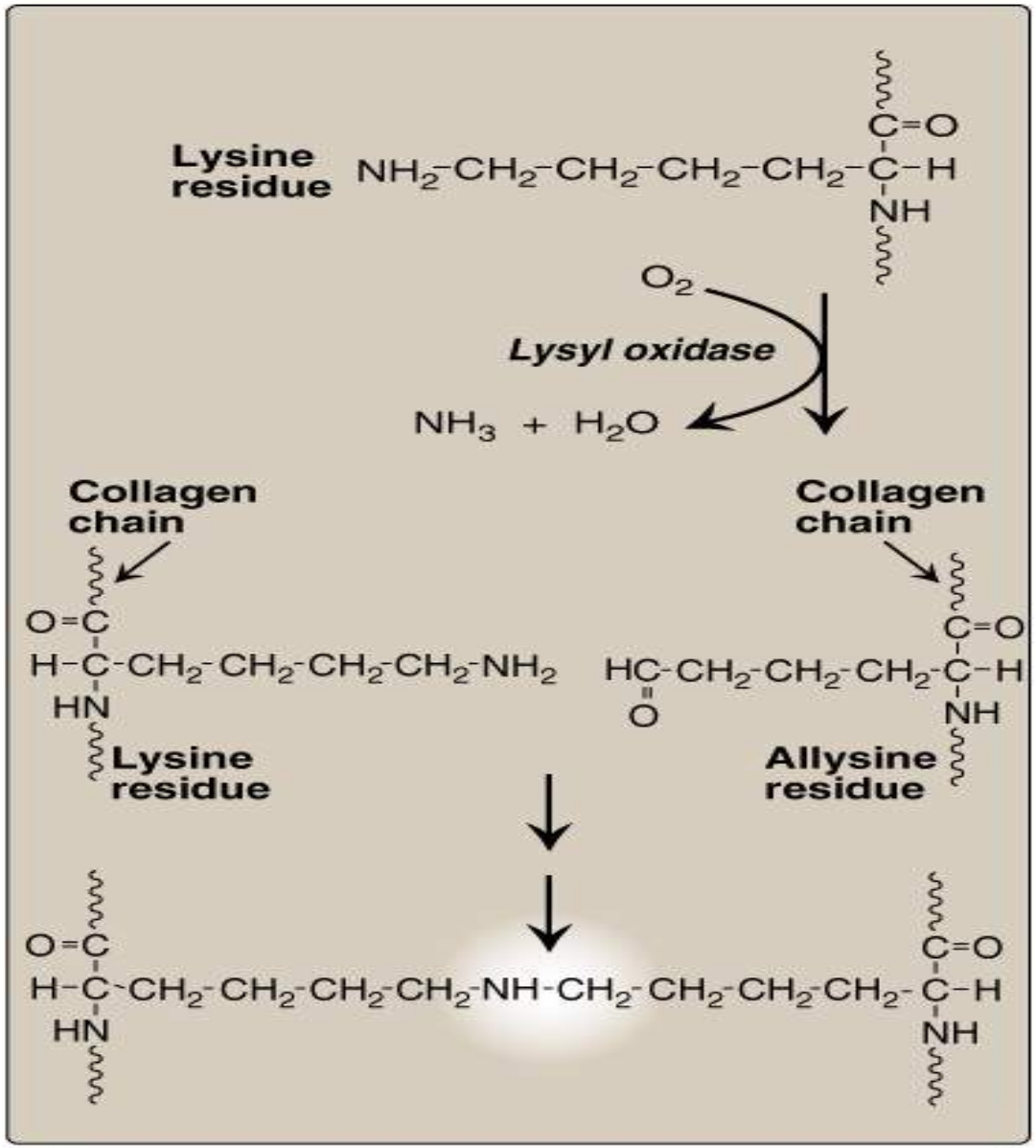


Figure 4.9

Formation of cross-links in collagen.

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Lecture 8

Contents:

- Degradation of collagen
- Collagen diseases
  (Osteogenesis Imperfecta)

D. Degradation of collagen

- **Normal collagens are highly stable molecules, having half-lives as long as several months.**

- **However, connective tissue is dynamic and is constantly being remodeled, often in response to growth or injury of the tissue.**

- **Breakdown of collagen fibrils is dependent on the proteolytic action of *collagenases*, which are part of a large family of matrix *metalloproteinases*.**

- **Type I collagen : the cleavage site is specific, generating three-quarter and one-quarter length fragments.**

- **These fragments are further degraded by other matrix proteinases to their constituent amino acids.**

E. Collagen diseases

- Defects in any one of the many steps in collagen fiber synthesis → genetic disease → inability of collagen to form fibers properly and → provide tissues with the needed tensile strength normally provided by collagen.

- **> 1000 mutations have been identified in 22 genes → coding for 12 of the collagen types.**

- **The following are examples of diseases that are the result of defective collagen synthesis.**

:Osteogenesis Imperfecta (OI)

- This disease, known as brittle bone syndrome, is also a heterogeneous group of inherited disorders distinguished by bones that easily bend and fracture *(Figure 4.11).*

- Retarded wound healing and a rotated and twisted spine → 'humped-back" appearance are common features of the disease.

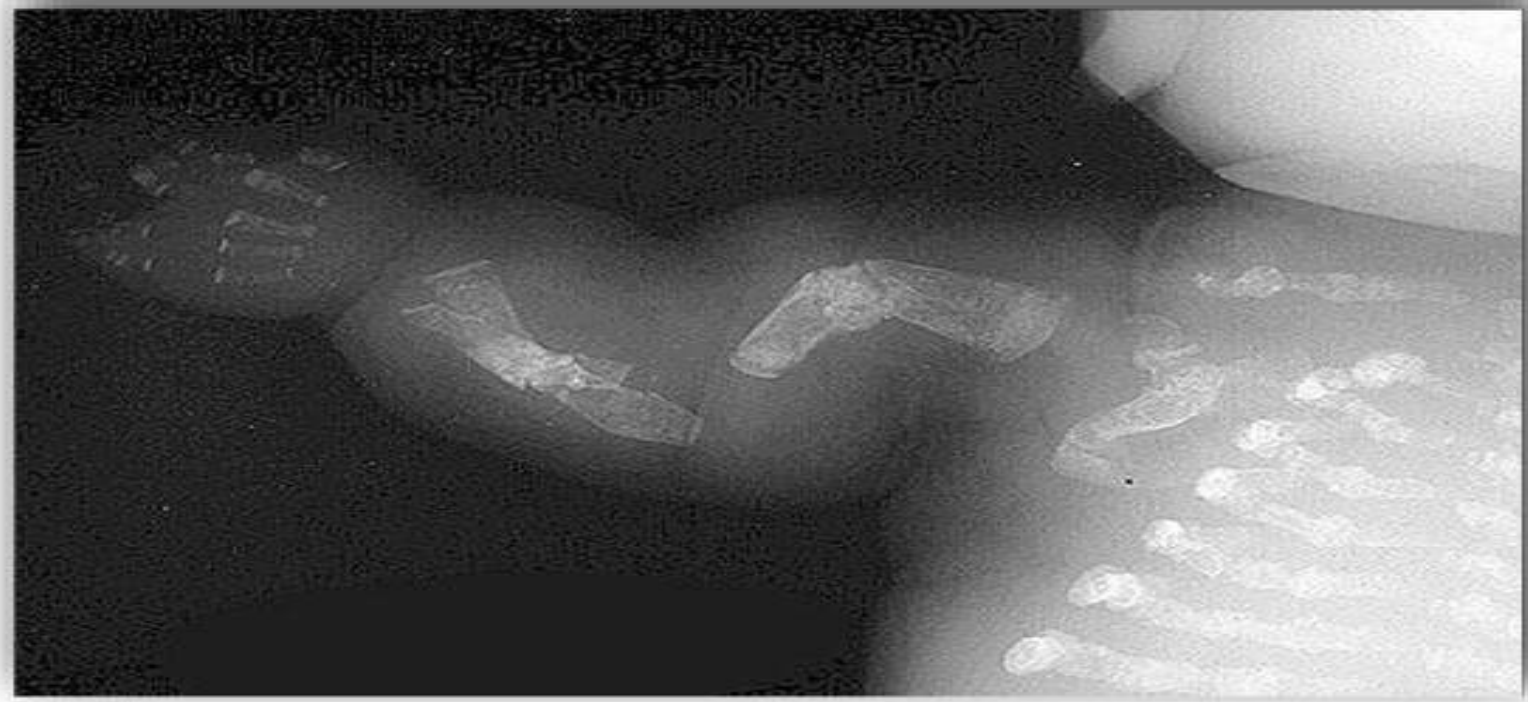


Figure 4.11

Lethal form of osteogenesis imperfecta in which the fractures appear in utero, as revealed by this radiograph of a stillborn fetus.

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- **Type I OI is called osteogenesis imperfecta tarda.**

- **This disease :**
  - **presents in early infancy.**
  - **fractures secondary to minor trauma.**
  - **may be suspected if prenatal ultrasound detects bowing or fractures of long bones.**

- **Type II OI, (osteogenesis imperfecta congenita):**
  - **more severe**
  - **patients die in utero or in the neonatal period of pulmonary hypoplasia.**
  - **Most patients with severe OI have mutations in the gene for either the pro 1- or pro2- α-chains of type I collagen.**

- The most common mutations → substitution of single amino acids with bulky side chains for the glycine residues that appear as every third amino acid in the triple helix.

- The structurally abnormal pro - α-chains → prevent folding of the protein into a triple-helical conformation.

Lecture 9

Contents:

- **Elastin**
  - **Structure of elastin**
  - **Role of α_1-antitrypsin in elastin degradation**

ELASTIN

- In contrast to collagen, which forms fibers that are tough and have high tensile strength, Elastin is a connective tissue protein with rubber-like properties.

- **Elastic fibers** composed of elastin and glycoprotein microfibrils are found in the **lungs**, the **walls of large arteries**, and **elastic ligaments**.

- They **can be stretched** to several times their normal length, but **recoil to their original shape** when the stretching force is relaxed.

A. Structure of elastin

- **Elastin is**
  - **an insoluble protein polymer**
  - **synthesized from a precursor, tropoelastin, *(a linear polypeptide composed of about 700 amino acids that are primarily small and nonpolar)* (e.g. glycine, alanine, and valine).**

- **Elastin is also,**
  - **rich in proline and lysine,**
  - **contains a little hydroxyproline**
  - **contains no hydroxylysine.**

- **Tropoelastin is secreted by the cell into the extracellular space.**

- **There, it interacts with specific glycoprotein microfibrils, such as fibrillin, which function as a scaffold onto which tropoelastin is deposited.**

- ***Mutations in the fibrillin gene are responsible for Marfan's syndrome***

- Some of the lysyl side chains of the tropoelastin polypeptides are oxidatively deaminated by lysyl oxidase → forming allysine residues.
- 3 of the allysyl side chains + one unaltered lysyl side chain from the same or neighboring polypeptides → form a desmosine cross-link. *(Figure 4.12).*
- This produces Elastin - an extensively interconnected, rubbery network that can stretch and bend in any direction when stressed → connective tissue elasticity

(Figure 4.13).

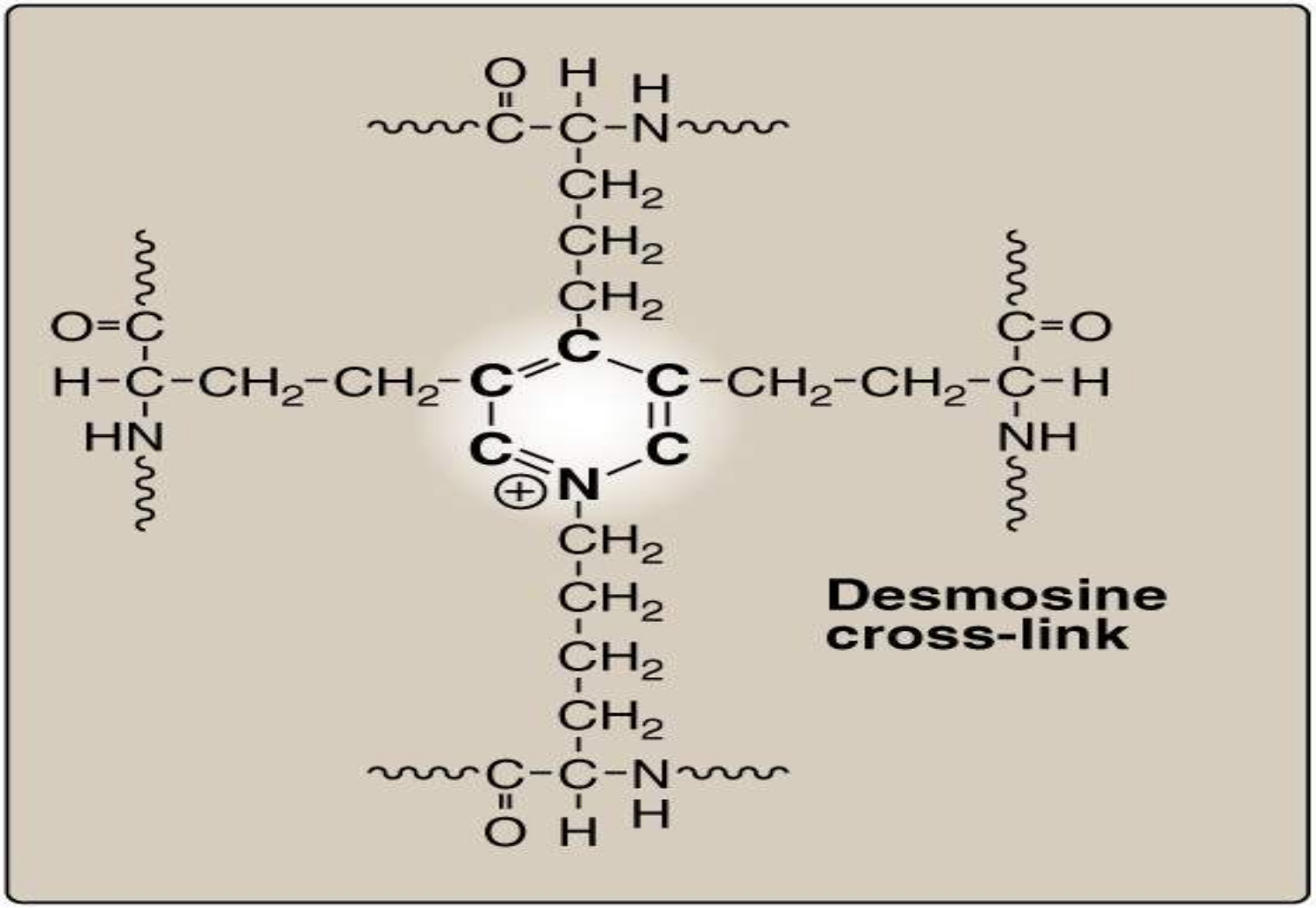


Figure 4.12

Desmosine cross-link in elastin.

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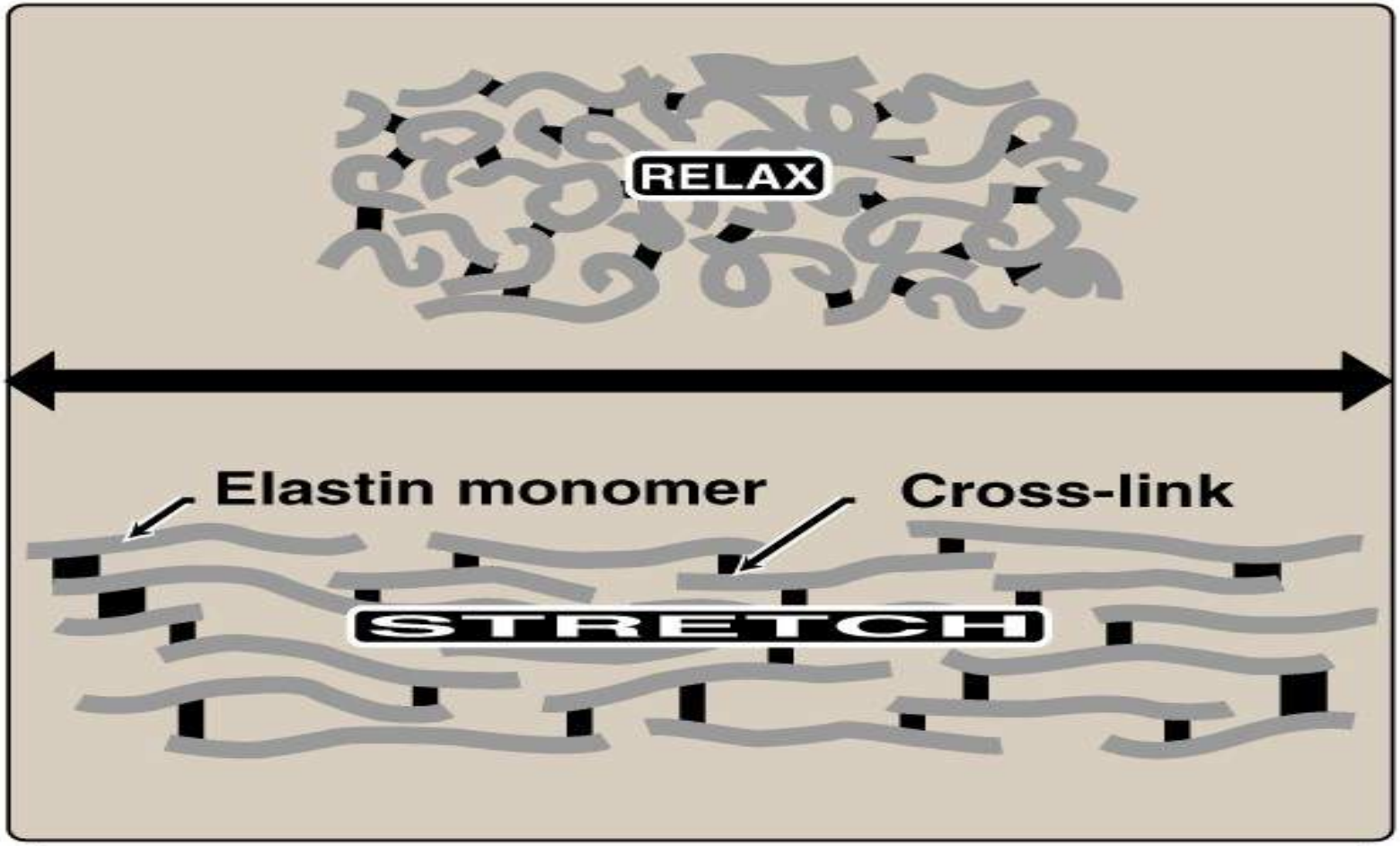


Figure 4.13
Elastin fibers in relaxed and stretched conformations.

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B. Role of α_1-antitrypsin in elastin degradation

1. α_1-antitrypsin:

- **Blood** and other body fluids contain a protein, α_1- antitrypsin (α_1-AT) or (α_1-antiproteinase) → **inhibits** a number of proteolytic enzymes (***proteases*** or ***proteinases***) → hydrolyze and destroy proteins.

- ***[The inhibitor was originally named α_1-antitrypsin because it inhibits the activity of trypsin (a proteolytic enzyme synthesized as trypsinogen by the pancreas]***

- **α_1- AT comprises > 90% of the α_1-globulin fraction of normal plasma.**

- α_1- AT has the important physiologic role of inhibiting neutrophil *elastase* - a powerful protease that is released into the extracellular space, and degrades elastin of alveolar walls, as well as other structural proteins in a variety of tissues *(Figure 4.14).*

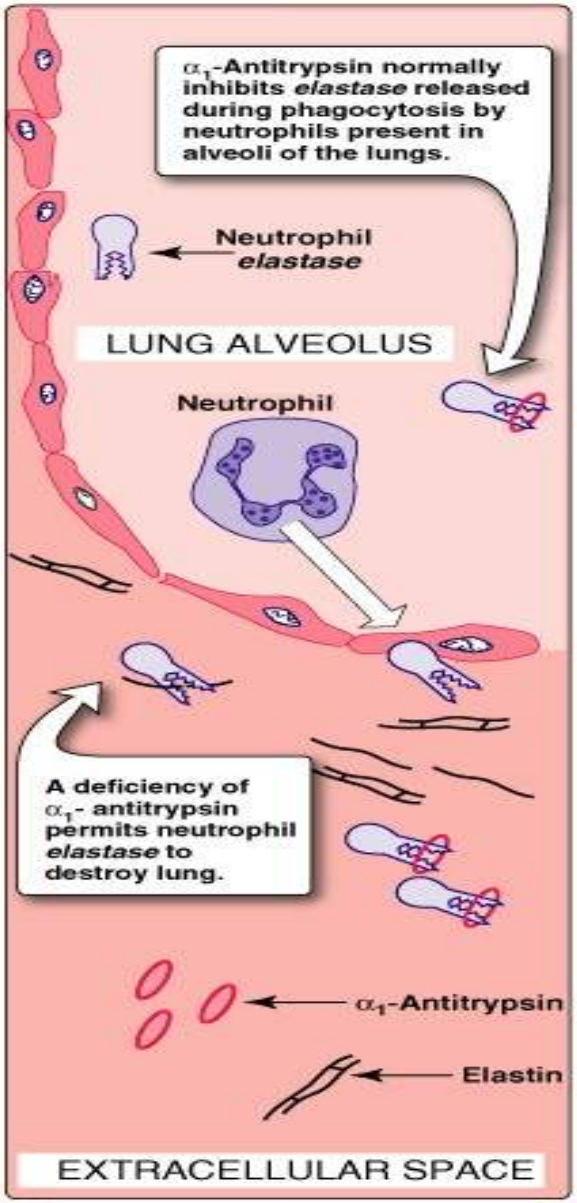


Figure 4.14
Destruction of alveolar tissue by elastase released from neutrophils.

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- **Most of the α_1-AT found in plasma is synthesized and secreted by the liver.**

- **The remainder is synthesized by several tissues, including monocytes and alveolar macrophages, which may be important in the prevention of local tissue injury by elastase.**

:Role of α_1-AT in the lungs .2

- **In the normal lung, the alveoli are chronically exposed to low levels of *neutrophil elastase* released from activated and degenerating neutrophils.**

- **This proteolytic activity can destroy the elastin in alveolar walls if unopposed by the inhibitory action of α_1-AT *(the most important inhibitor of neutrophil elastase)***

(Figure 4.14).

- **Because lung tissue cannot regenerate, emphysema results from the destruction of the connective tissue of alveolar walls.**

Emphysema resulting from α_1-AT .3 :deficiency

- In USA, inherited defects in α_1-AT → ~ 2-5% of patients having emphysema.
- A number of different mutations in the α_1-AT gene → deficiency of this protein, but one single purine base mutation (GAG → AAG, → substitution of lysine for glutamic acid at position 342 of the protein) is clinically the most widespread.

- An individual must inherit 2 abnormal α_1- AT alleles to be at risk for the development of emphysema.

- In a heterozygote (with one normal and one defective gene) the levels of α_1-AT are sufficient to protect the alveoli from damage.

- ***A specific α_1-AT methionine is required for the binding of the inhibitor to its target proteases.***
- ***Smoking → oxidation and subsequent inactivation of that methionine residue, → render the inhibitor powerless to neutralize elastase.***
- ***Smokers with α_1-AT deficiency, have ⇈ rate of lung destruction → poorer survival rate than nonsmokers with the deficiency.***

- **The deficiency of *elastase inhibitor* can be reversed by weekly IV administration of α_1-AT.**

- **The α_1-AT diffuses from the blood into the lung, where it reaches therapeutic levels in the fluid surrounding the lung epithelial cells.**

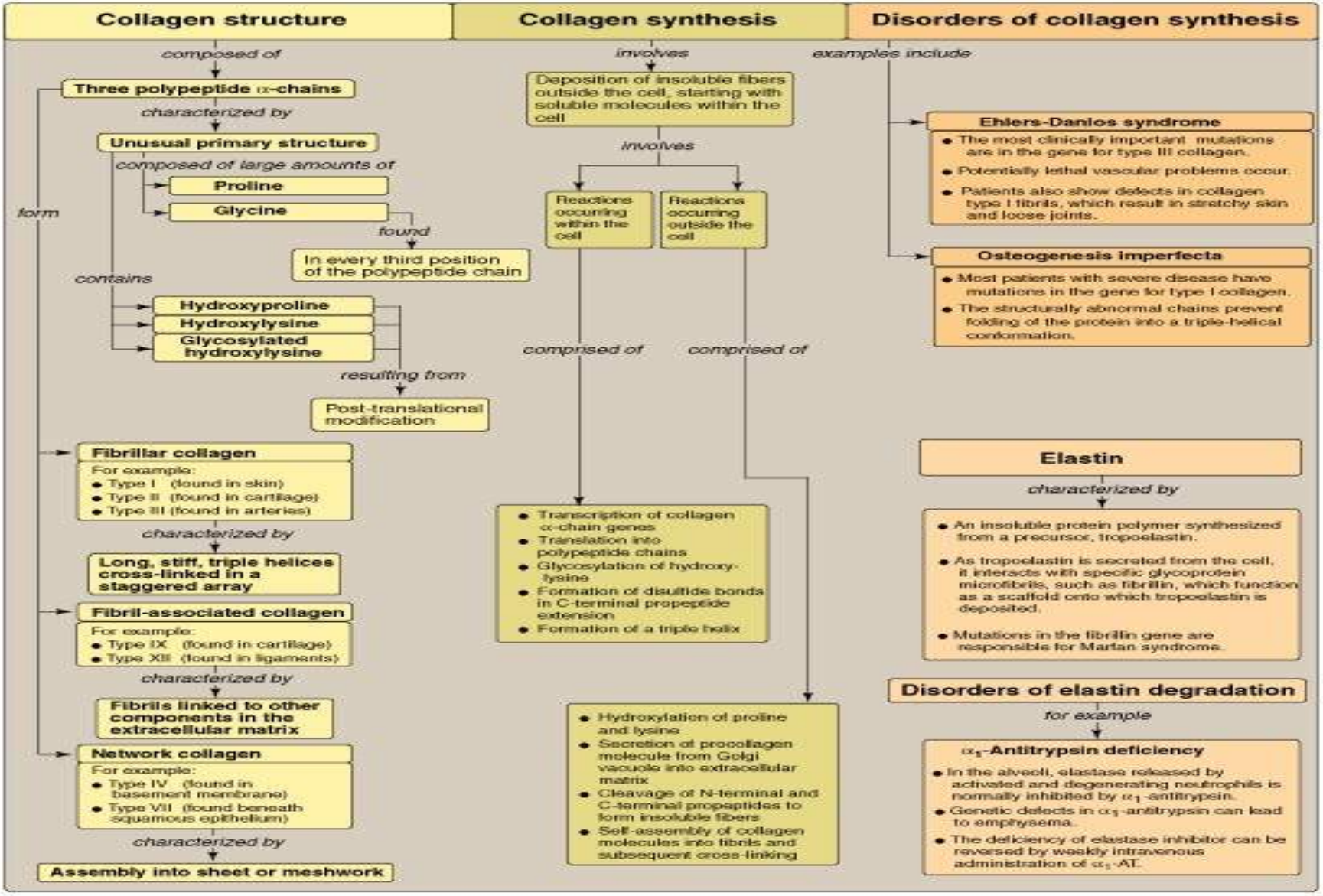


Figure 4.15
Key concept map for the fibrous proteins, collagen, and elastin.

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THE END